

模式识别与机器学习

授课教师：杨卫东 胡静

L5-2

1



第七讲 线性支撑向量机 (*Linear Support Vector Machine*)



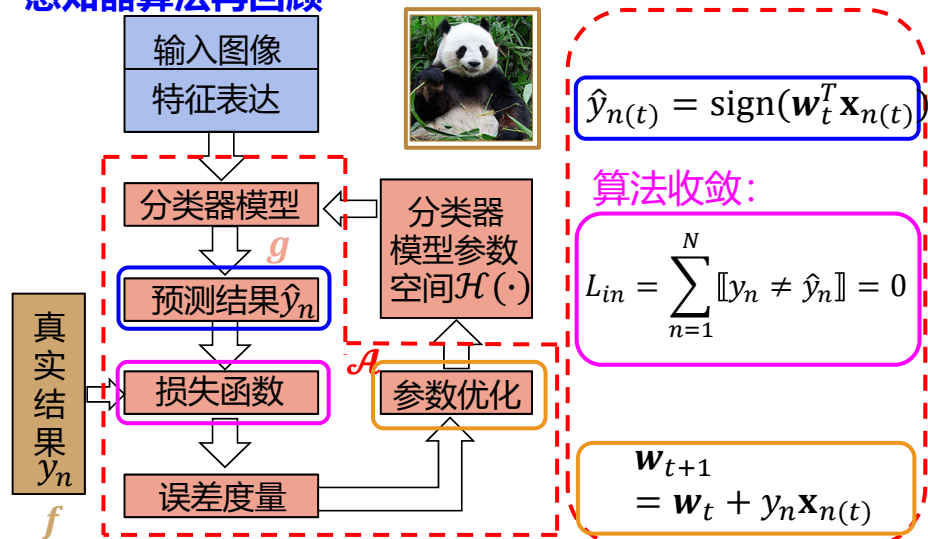
7.1 最大间隔分类面 (*Large-Margin Separating Hyperplane*)

7.2 标准的最大间隔问题 (*Standard Large-Margin Problem*)

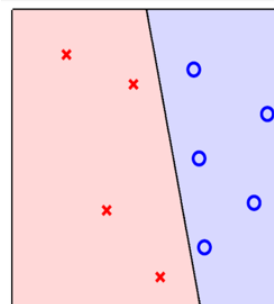
7.3 支撑向量机 (*Support Vector Machine*)

7.1 最大间隔分类面

感知器算法再回顾



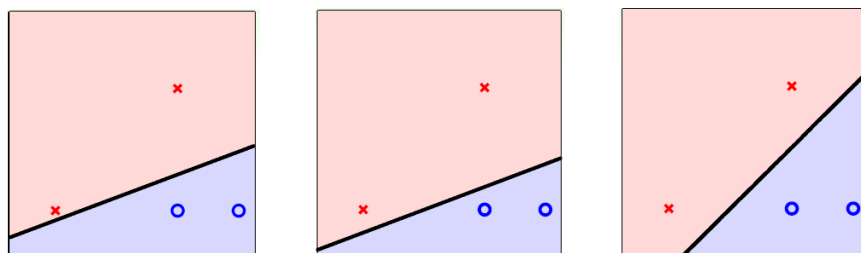
线性可分



- 设置初始分类面 (权重) $\mathbf{w}_0$
- 如果有样本分错, 就修正权重

7.1 最大间隔分类面

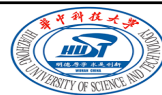
哪一个分类面最佳?



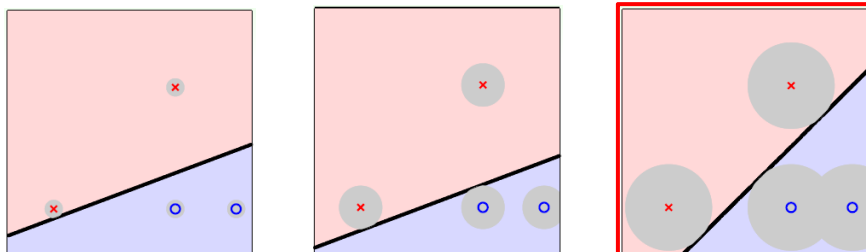
- 感知器算法能找出吗?
- VC bound?

$$E_{out}(\mathbf{w}) \leq \underbrace{E_{in}(\mathbf{w})}_0 + \underbrace{\Omega(\mathcal{H})}_{d_{VC}=d+1}$$

7.1 最大间隔分类面



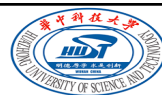
为什么会认为最右边的分类面最佳?



假设 $\mathbf{x} \approx \mathbf{x}_n + \Delta \mathbf{x}_n$, $\Delta \mathbf{x}_n \sim N(\mathbf{x}_n, \sigma_n)$

$\mathbf{x}_n$ 离分类面越远 $\Leftrightarrow$ 容许 σ_n 越大, 噪声的容忍度越大 $\Leftrightarrow$ 不易出现过拟合
 $\Leftrightarrow$ 更鲁棒分类面 $\Leftrightarrow$ 对噪声的容忍度更大 $\Leftrightarrow$ 离分类面最近的 $\mathbf{x}_n$ 到分类面距离最大
 最右边的最佳---因为离分类面最近的 $\mathbf{x}_n$ 到分类面距离最大, 对噪声最鲁棒

7.1 最大间隔分类面



对噪声鲁棒的分类面

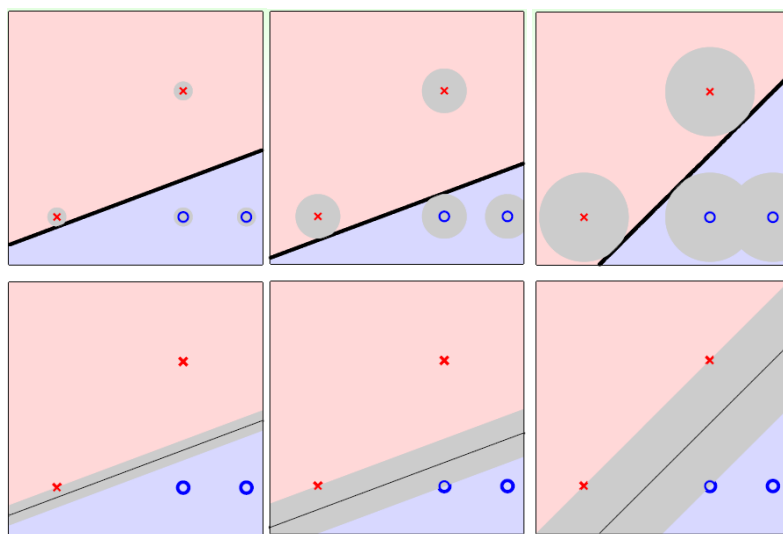


分类面到两边样本的距离要大



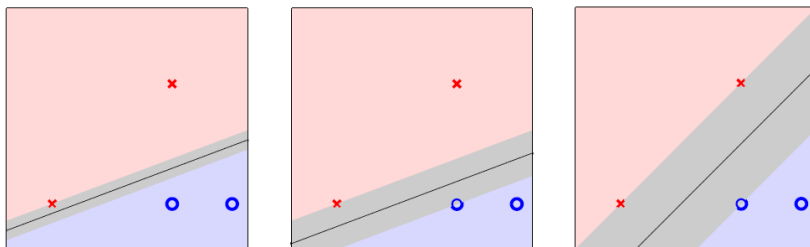
“胖胖”的分类面

算法的目的是如何找到
“胖胖”的分类面



7.1 最大间隔分类面

“胖胖”的分类面



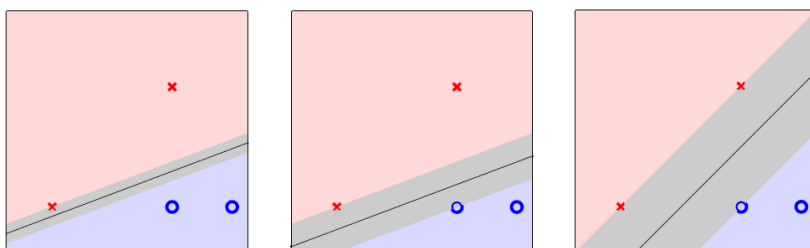
$$\begin{aligned} & \max_{\mathbf{w}} \quad \text{fatness}(\mathbf{w}) \\ & \text{subject to} \quad \mathbf{w} \text{ classifies every } (\mathbf{x}_n, y_n) \text{ correctly} \\ & \quad \text{fatness}(\mathbf{w}) = \min_{n=1, \dots, N} \text{distance}(\mathbf{x}_n, \mathbf{w}) \end{aligned}$$

7.1 最大间隔分类面

“胖胖”的分类面



最大间隔分类面



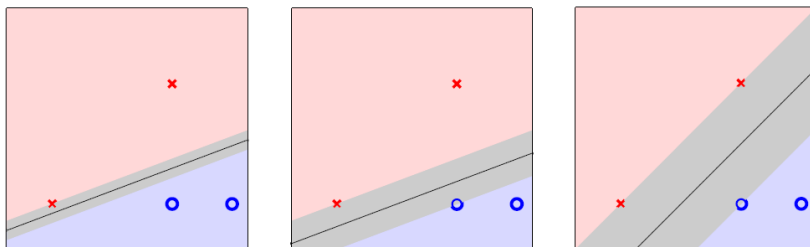
$$\begin{aligned} & \max_{\mathbf{w}} \quad \text{margin}(\mathbf{w}) \\ & \text{subject to} \quad \mathbf{w} \text{ classifies every } (\mathbf{x}_n, y_n) \text{ correctly} \\ & \quad \text{margin}(\mathbf{w}) = \min_{n=1, \dots, N} \text{distance}(\mathbf{x}_n, \mathbf{w}) \end{aligned}$$

7.1 最大间隔分类面

“胖胖”的分类面



最大间隔分类面



所有样本正确分类



$$y_n = \text{sign}(\mathbf{w}^T \mathbf{x}_n)$$

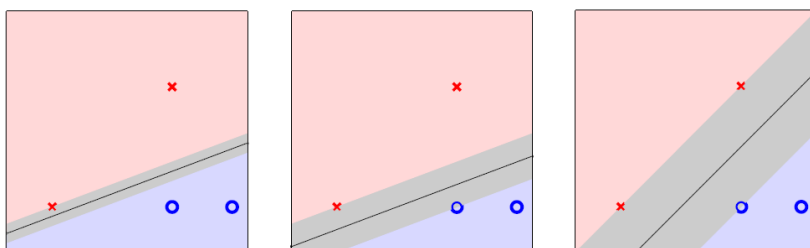
$$\begin{aligned} & \max_{\mathbf{w}} \quad \text{margin}(\mathbf{w}) \\ & \text{subject to} \quad \mathbf{w} \text{ classifies every } (\mathbf{x}_n, y_n) \text{ correctly} \\ & \quad \text{margin}(\mathbf{w}) = \min_{n=1, \dots, N} \text{distance}(\mathbf{x}_n, \mathbf{w}) \end{aligned}$$

7.1 最大间隔分类面

“胖胖”的分类面



最大间隔分类面



所有样本正确分类

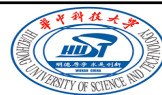


$$y_n = \text{sign}(\mathbf{w}^T \mathbf{x}_n)$$

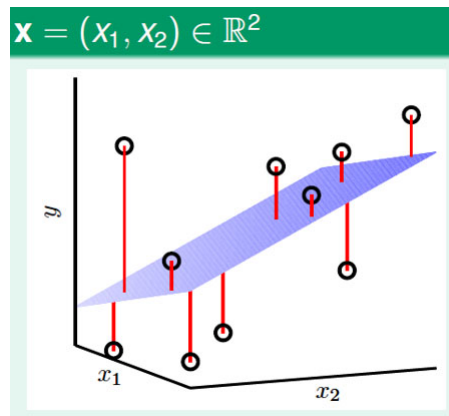
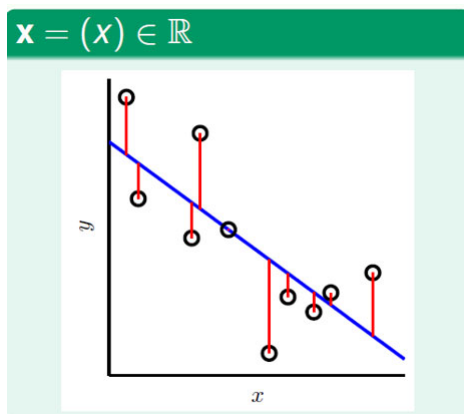
$$\begin{aligned} & \max_{\mathbf{w}} \quad \text{margin}(\mathbf{w}) \\ & \text{subject to} \quad \text{every } y_n \mathbf{w}^T \mathbf{x}_n > 0 \\ & \quad \text{margin}(\mathbf{w}) = \min_{n=1, \dots, N} \text{distance}(\mathbf{x}_n, \mathbf{w}) \end{aligned}$$

算法的目的是：如何找到“最大间隔”的分类面 (*find largest-margin separating hyperplane*)

3.1 线性回归问题（引）

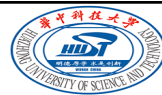


线性回归示例



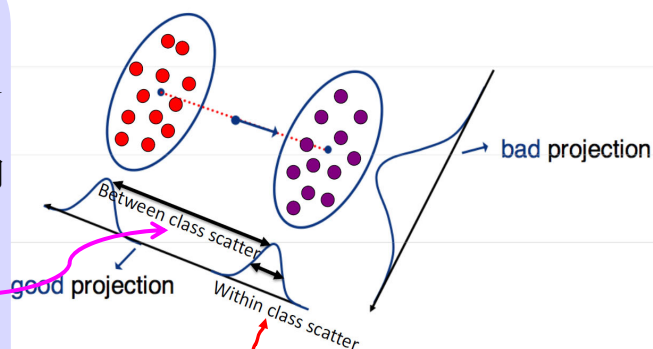
回归问题: $y \in \mathbb{R}$

4.1 Fisher线性判别动机（引）



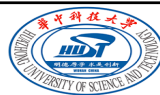
Fisher线性判别的目的:

- 在尽可能保留类别可区分性的前提下实现维数减少
(Perform dimensionality reduction “while preserving as much of the class discriminatory information as possible”.)
- 找到让类别最好区分的投影方向
(Seeks to find directions along which the classes are best separated.)
- 同时考虑类内散布和类间散布
(Takes into consideration the scatter within-classes but also the scatter between-classes.)



Slide inspired by N. Vasconcelos

第七讲 线性支撑向量机 (Linear Support Vector Machine)

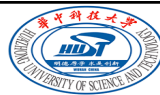


7.1 最大间隔分类面 (Large-Margin Separating Hyperplane)

7.2 标准的最大间隔问题 (Standard Large-Margin Problem)

7.3 支撑向量机 (Support Vector Machine)

7.2 标准的最大间隔问题



样本到分类面的距离

$$\begin{aligned} & \max_{\mathbf{w}} \quad \text{margin}(\mathbf{w}) \\ & \text{Subject to} \quad \text{every } y_n \mathbf{w}^T \mathbf{x}_n > 0 \\ & \quad \text{margin}(\mathbf{w}) = \min_{n=1, \dots, N} \text{distance}(\mathbf{x}_n, \mathbf{w}) \end{aligned}$$

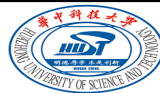
为了便于了解分类器特性，支撑向量机的分析过程中样本向量不做增广

$$\begin{aligned} b &= w_0 \\ \begin{bmatrix} | \\ \mathbf{w} \\ | \end{bmatrix} &= \begin{bmatrix} w_1 \\ \vdots \\ w_d \end{bmatrix} ; \quad \begin{bmatrix} | \\ \mathbf{x} \\ | \end{bmatrix} = \begin{bmatrix} x_1 \\ \vdots \\ x_d \end{bmatrix} \end{aligned}$$

~~$x_0 = 1$~~

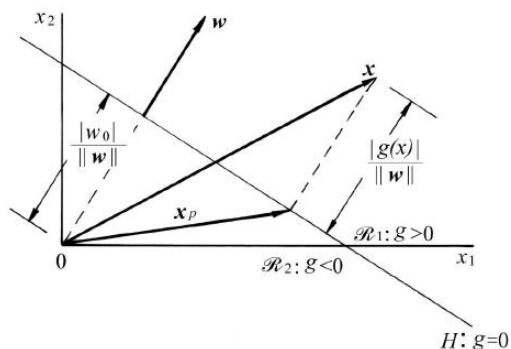
$$h(\mathbf{x}) = \text{sign}(\mathbf{w}^T \mathbf{x} + b)$$

7.2 标准的最大间隔问题



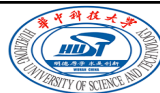
样本到分类面的距离

$$\begin{aligned} & \max_w \quad \text{margin}(w) \\ \text{Subject to} \quad & \text{every } y_n(w^T x_n + b) > 0 \\ & \text{margin}(w) = \min_{n=1, \dots, N} \text{distance}(x_n, w) \end{aligned}$$



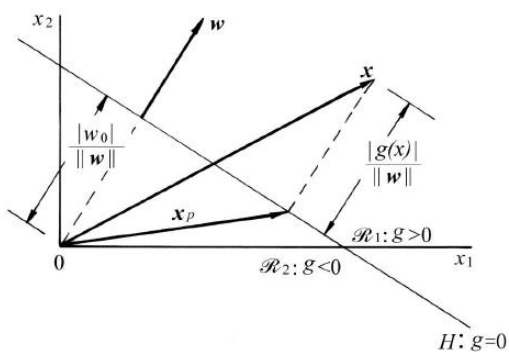
$$\begin{aligned} g(x) &= w^T x + w_0 \\ &= w^T \left(x_p + r \frac{w}{\|w\|} \right) + w_0 \\ &= w^T x_p + w_0 + r \frac{w^T w}{\|w\|} \\ &= r \|w\| \end{aligned}$$

7.2 标准的最大间隔问题



样本到分类面的距离

$$\begin{aligned} & \max_w \quad \text{margin}(w) \\ \text{Subject to} \quad & \text{every } y_n(w^T x_n + b) > 0 \\ & \text{margin}(w) = \min_{n=1, \dots, N} \text{distance}(x_n, w) \end{aligned}$$



$$\begin{aligned} g(x) &= w^T x + b \\ &= w^T \left(x_p + r \frac{w}{\|w\|} \right) + b \\ &= w^T x_p + b + r \frac{w^T w}{\|w\|} \\ &= r \|w\| \end{aligned}$$

$$|r| = \frac{|g(x)|}{\|w\|} \Rightarrow \text{distance}(x_n, w) = \frac{1}{\|w\|} |w^T x + b|$$

7.2 标准的最大间隔问题

样本到分类面的距离

$$\text{distance}(\mathbf{x}_n, \mathbf{w}) = \frac{1}{\|\mathbf{w}\|} |\mathbf{w}^T \mathbf{x}_n + b|$$

$$\because \text{every } y_n(\mathbf{w}^T \mathbf{x}_n + b) > 0$$

$$\text{distance}(\mathbf{x}_n, \mathbf{w}) = \frac{1}{\|\mathbf{w}\|} y_n(\mathbf{w}^T \mathbf{x}_n + b)$$

$$\begin{aligned} & \max_{\mathbf{w}} \quad \text{margin}(\mathbf{w}) \\ & \text{Subject to} \quad \text{every } y_n(\mathbf{w}^T \mathbf{x}_n + b) > 0 \\ & \text{margin}(\mathbf{w}) = \min_{n=1, \dots, N} \text{distance}(\mathbf{x}_n, \mathbf{w}) \end{aligned}$$



$$\begin{aligned} & \max_{\mathbf{w}} \quad \text{margin}(\mathbf{w}) \\ & \text{Subject to} \quad \text{every } y_n(\mathbf{w}^T \mathbf{x}_n + b) > 0 \end{aligned}$$

$$\text{margin}(\mathbf{w}) = \min_{n=1, \dots, N} \frac{1}{\|\mathbf{w}\|} y_n(\mathbf{w}^T \mathbf{x}_n + b)$$

7.2 标准的最大间隔问题

分类面 :

$$\mathbf{w}^T \mathbf{x}_n + b = 0$$

$$\begin{aligned} & \max_{\mathbf{w}} \quad \text{margin}(\mathbf{w}) \\ & \text{Subject to} \quad \text{every } y_n(\mathbf{w}^T \mathbf{x}_n + b) > 0 \end{aligned}$$

$$\text{margin}(\mathbf{w}) = \min_{n=1, \dots, N} \frac{1}{\|\mathbf{w}\|} y_n(\mathbf{w}^T \mathbf{x}_n + b)$$

7.2 标准的最大间隔问题

分类面的尺度缩放:

$$\mathbf{w}^T \mathbf{x}_n + b = 0$$

$$3\mathbf{w}^T \mathbf{x}_n + 3b = 0$$

$$\min_{n=1,\dots,N} y_n(\mathbf{w}^T \mathbf{x}_n + b) = 1$$



$$\text{margin}(\mathbf{w}) = \frac{1}{\|\mathbf{w}\|}$$

$$\begin{array}{ll} \max_{\mathbf{w}} & \text{margin}(\mathbf{w}) \\ \text{Subject to} & \text{every } y_n(\mathbf{w}^T \mathbf{x}_n + b) > 0 \end{array}$$

$$\text{margin}(\mathbf{w}) = \min_{n=1,\dots,N} \frac{1}{\|\mathbf{w}\|} y_n(\mathbf{w}^T \mathbf{x}_n + b)$$



$$\begin{array}{ll} \max_{\mathbf{w}} & \frac{1}{\|\mathbf{w}\|} \\ \text{Subject to} & \text{every } y_n(\mathbf{w}^T \mathbf{x}_n + b) > 0 \\ & \min_{n=1,\dots,N} y_n(\mathbf{w}^T \mathbf{x}_n + b) = 1 \end{array}$$

7.2 标准的最大间隔问题

分类面的尺度缩放:

$$\mathbf{w}^T \mathbf{x}_n + b = 0$$

$$3\mathbf{w}^T \mathbf{x}_n + 3b = 0$$

$$\min_{n=1,\dots,N} y_n(\mathbf{w}^T \mathbf{x}_n + b) = 1$$



$$\text{margin}(\mathbf{w}) = \frac{1}{\|\mathbf{w}\|}$$

$$\begin{array}{ll} \max_{\mathbf{w}} & \text{margin}(\mathbf{w}) \\ \text{Subject to} & \text{every } y_n(\mathbf{w}^T \mathbf{x}_n + b) > 0 \end{array}$$

$$\text{margin}(\mathbf{w}) = \min_{n=1,\dots,N} \frac{1}{\|\mathbf{w}\|} y_n(\mathbf{w}^T \mathbf{x}_n + b)$$



$$\begin{array}{ll} \max_{\mathbf{w}} & \frac{1}{\|\mathbf{w}\|} \\ \text{Subject to} & \text{every } y_n(\mathbf{w}^T \mathbf{x}_n + b) > 0 \\ & \min_{n=1,\dots,N} y_n(\mathbf{w}^T \mathbf{x}_n + b) = 1 \end{array}$$

7.2 标准的最大间隔问题

标准的最大间隔问题：

原始约束条件下的值域
 $\min_{n=1,\dots,N} y_n(\mathbf{w}^T \mathbf{x}_n + b) = 1$

$$\max_{\mathbf{w}} \frac{1}{\|\mathbf{w}\|}$$

$$\text{Subject to } \min_{n=1,\dots,N} y_n(\mathbf{w}^T \mathbf{x}_n + b) = 1$$

7.2 标准的最大间隔问题

标准的最大间隔问题：

条件松弛后的值域
 $y_n(\mathbf{w}^T \mathbf{x}_n + b) \geq 1, \text{ for all } n$

原始约束条件下的值域
 $\min_{n=1,\dots,N} y_n(\mathbf{w}^T \mathbf{x}_n + b) = 1$

条件松弛后的解
 仍然在紫色值域

$$\max_{\mathbf{w}} \frac{1}{\|\mathbf{w}\|}$$

$$\text{Subject to } \min_{n=1,\dots,N} y_n(\mathbf{w}^T \mathbf{x}_n + b) = 1$$

如果 $(\mathbf{w}^*, b^*)$ 位于蓝色值域，即对所有样本：

$$y_n(\mathbf{w}^{*T} \mathbf{x}_n + b^*) \geq 1.6$$

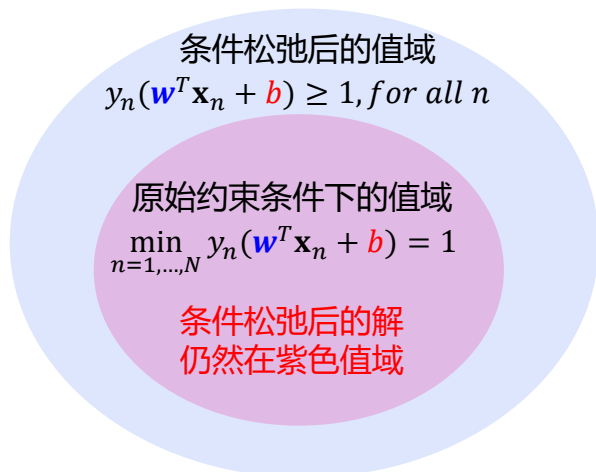
根据分类面 $(\mathbf{w}^*, b^*)$ 取值的尺度不变：

$$y_n\left(\frac{\mathbf{w}^{*T}}{1.6} \mathbf{x}_n + \frac{b^*}{1.6}\right) \geq 1$$

$(\frac{\mathbf{w}^{*T}}{1.6}, \frac{b^*}{1.6})$ 为最佳解 ➡ 矛盾!!!

7.2 标准的最大间隔问题

标准的最大间隔问题：



$$\max_{\mathbf{w}} \quad \frac{1}{\|\mathbf{w}\|}$$

$$\text{Subject to} \quad \min_{n=1, \dots, N} y_n(\mathbf{w}^T \mathbf{x}_n + b) = 1$$

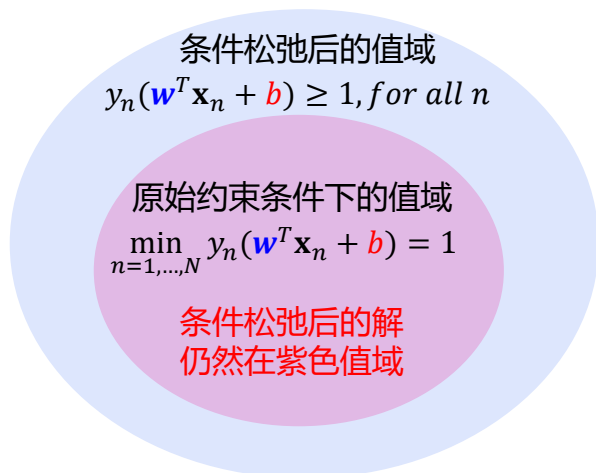


$$\max_{\mathbf{w}} \quad \frac{1}{\|\mathbf{w}\|}$$

$$\text{Subject to} \quad y_n(\mathbf{w}^T \mathbf{x}_n + b) \geq 1, \text{ for all } n$$

7.2 标准的最大间隔问题

标准的最大间隔问题：



$$\max_{\mathbf{w}} \quad \frac{1}{\|\mathbf{w}\|}$$

$$\text{Subject to} \quad \min_{n=1, \dots, N} y_n(\mathbf{w}^T \mathbf{x}_n + b) = 1$$

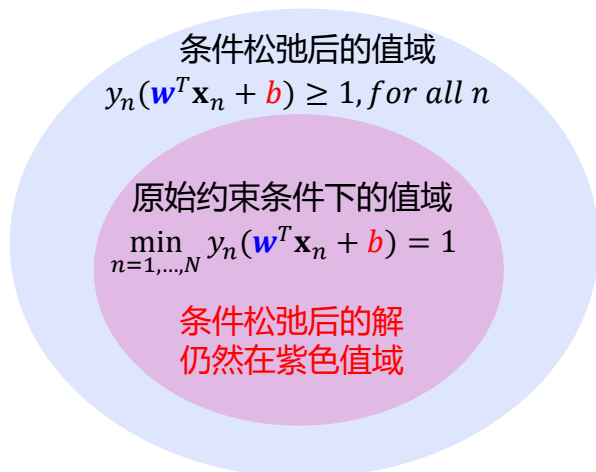


$$\min_{\mathbf{w}} \quad \|\mathbf{w}\|$$

$$\text{Subject to} \quad y_n(\mathbf{w}^T \mathbf{x}_n + b) \geq 1, \text{ for all } n$$

7.2 标准的最大间隔问题

标准的最大间隔问题：



$$\max_{\mathbf{w}} \quad \frac{1}{\|\mathbf{w}\|}$$

$$\text{Subject to} \quad \min_{n=1, \dots, N} y_n(\mathbf{w}^T \mathbf{x}_n + b) = 1$$

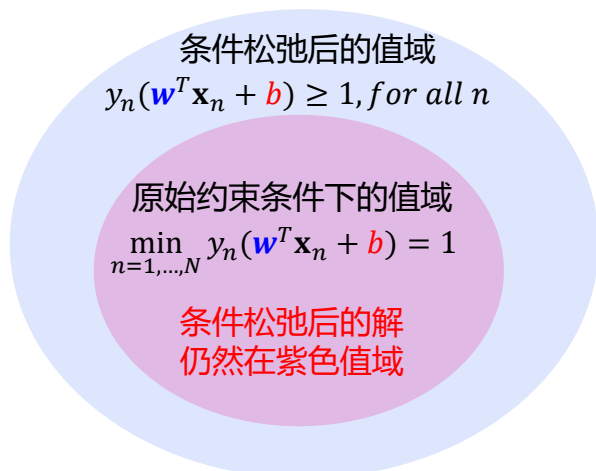


$$\min_{\mathbf{w}} \quad \sqrt{\mathbf{w}^T \mathbf{w}}$$

$$\text{Subject to} \quad y_n(\mathbf{w}^T \mathbf{x}_n + b) \geq 1, \text{ for all } n$$

7.2 标准的最大间隔问题

标准的最大间隔问题：



$$\max_{\mathbf{w}} \quad \frac{1}{\|\mathbf{w}\|}$$

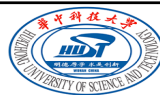
$$\text{Subject to} \quad \min_{n=1, \dots, N} y_n(\mathbf{w}^T \mathbf{x}_n + b) = 1$$



$$\min_{\mathbf{w}} \quad \frac{1}{2} \mathbf{w}^T \mathbf{w}$$

$$\text{Subject to} \quad y_n(\mathbf{w}^T \mathbf{x}_n + b) \geq 1, \text{ for all } n$$

第七讲 线性支撑向量机 (Linear Support Vector Machine)



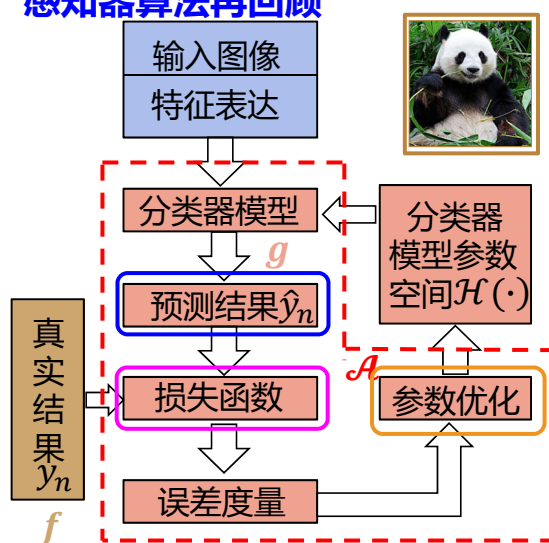
7.1 最大间隔分类面 (Large-Margin Separating Hyperplane)

7.2 标准的最大间隔问题 (Standard Large-Margin Problem)

7.3 支撑向量机 (Support Vector Machine)

7.3 支撑向量机

感知器算法再回顾



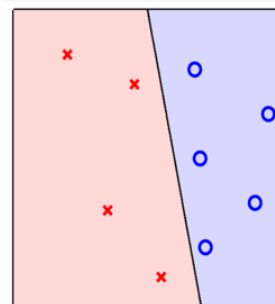
$$\hat{y}_{n(t)} = \text{sign}(\mathbf{w}_t^T \mathbf{x}_{n(t)})$$

算法收敛:

$$L_{in} = \sum_{n=1}^N \mathbb{I}[y_n \neq \hat{y}_n] = 0$$

$$\mathbf{w}_{t+1} = \mathbf{w}_t + y_n \mathbf{x}_{n(t)}$$

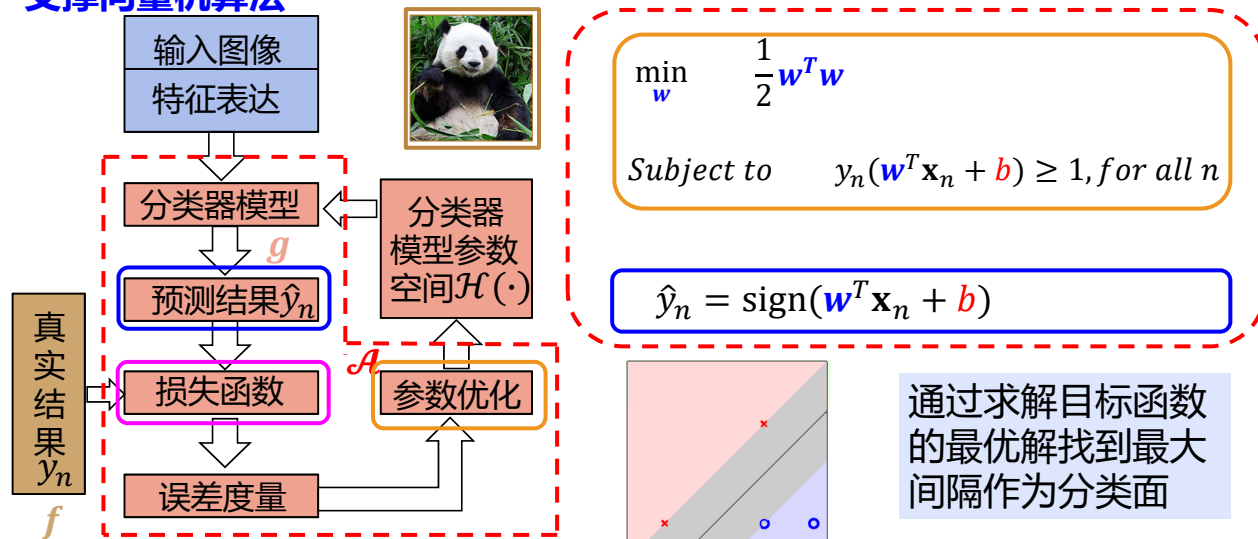
线性可分



- 设置初始分类面 (权重) $\mathbf{w}_0$
- 如果有样本分错, 就修正权重

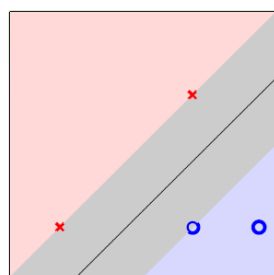
7.3 支撑向量机

支撑向量机算法



7.3 支撑向量机

最大间隔面的求解示例：



$$\mathbf{x} = \begin{bmatrix} 0 & 0 \\ 2 & 2 \\ 2 & 0 \\ 3 & 0 \end{bmatrix} \quad \mathbf{y} = \begin{bmatrix} -1 \\ -1 \\ +1 \\ +1 \end{bmatrix}$$

$$\min_w \frac{1}{2} w^T w$$

$$\text{Subject to } y_n(w^T x_n + b) \geq 1, \text{ for all } n$$

由约束条件可得：

$$\begin{cases} -b \geq 1 & (1) \\ -2w_1 - 2w_2 - b \geq 1 & (2) \\ 2w_1 + b \geq 1 & (3) \\ 3w_1 + b \geq 1 & (4) \end{cases}$$

对约束条件计算可得：

$$\begin{aligned} (1) + (3) &\Rightarrow w_1 \geq +1 \\ (2) + (3) &\Rightarrow w_2 \leq -1 \end{aligned}$$

由目标函数 $\frac{1}{2} w^T w$ 取最小值可得：

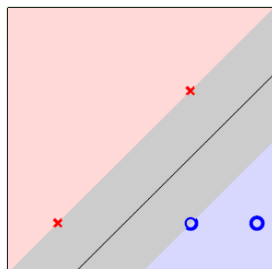
$$w_1 = 1, w_2 = -1$$

代入约束条件：

$$b \leq -1, \quad b \leq -1, \quad b \geq -1, \quad b \geq -2;$$

7.3 支撑向量机

最大间隔面的求解示例：



$$\mathbf{X} = \begin{bmatrix} 0 & 0 \\ 2 & 2 \\ 2 & 0 \\ 3 & 0 \end{bmatrix} \quad \mathbf{y} = \begin{bmatrix} -1 \\ -1 \\ +1 \\ +1 \end{bmatrix}$$

$$\min_{\mathbf{w}} \quad \frac{1}{2} \mathbf{w}^T \mathbf{w}$$

$$\text{Subject to} \quad y_n(\mathbf{w}^T \mathbf{x}_n + b) \geq 1, \text{ for all } n$$

由约束条件可得：

$$\begin{cases} -b \geq 1 & (1) \\ -2w_1 - 2w_2 - b \geq 1 & (2) \\ 2w_1 + b \geq 1 & (3) \\ 3w_1 + b \geq 1 & (4) \end{cases}$$

对约束条件计算可得：

$$(1) + (3) \Rightarrow w_1 \geq +1$$

$$(2) + (3) \Rightarrow w_2 \leq -1$$

由目标函数 $\frac{1}{2} \mathbf{w}^T \mathbf{w}$ 取最小值可得：

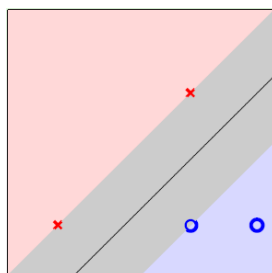
$$w_1 = 1, w_2 = -1$$

代入约束条件：

$$b \leq -1, \quad b \leq -1, \quad b \geq -1, \quad b \geq -2;$$

7.3 支撑向量机

最大间隔面的求解示例：



$$\mathbf{X} = \begin{bmatrix} 0 & 0 \\ 2 & 2 \\ 2 & 0 \\ 3 & 0 \end{bmatrix} \quad \mathbf{y} = \begin{bmatrix} -1 \\ -1 \\ +1 \\ +1 \end{bmatrix}$$

$$\min_{\mathbf{w}} \quad \frac{1}{2} \mathbf{w}^T \mathbf{w}$$

$$\text{Subject to} \quad y_n(\mathbf{w}^T \mathbf{x}_n + b) \geq 1, \text{ for all } n$$

由约束条件可得：

$$\begin{cases} -b \geq 1 & (1) \\ -2w_1 - 2w_2 - b \geq 1 & (2) \\ 2w_1 + b \geq 1 & (3) \\ 3w_1 + b \geq 1 & (4) \end{cases}$$

对约束条件计算可得：

$$(1) + (3) \Rightarrow w_1 \geq +1$$

$$(2) + (3) \Rightarrow w_2 \leq -1$$

由目标函数 $\frac{1}{2} \mathbf{w}^T \mathbf{w}$ 取最小值可得：

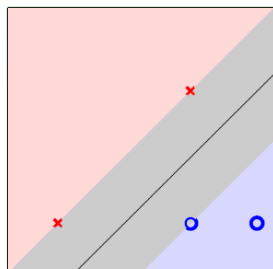
$$w_1 = 1, w_2 = -1$$

代入约束条件： $b = -1$

$$\hat{y}_n = \text{sign}(\mathbf{w}^T \mathbf{x}_n + b)$$

7.3 支撑向量机

最大间隔面的求解示例：



$$\mathbf{X} = \begin{bmatrix} 0 & 0 \\ 2 & 2 \\ 2 & 0 \\ 3 & 0 \end{bmatrix} \quad \mathbf{y} = \begin{bmatrix} -1 \\ -1 \\ +1 \\ +1 \end{bmatrix}$$

$$\min_{\mathbf{w}} \quad \frac{1}{2} \mathbf{w}^T \mathbf{w}$$

$$\text{Subject to} \quad y_n(\mathbf{w}^T \mathbf{x}_n + b) \geq 1, \text{ for all } n$$

由约束条件可得：

$$\begin{cases} -b \geq 1 & (1) \\ -2w_1 - 2w_2 - b \geq 1 & (2) \\ 2w_1 + b \geq 1 & (3) \\ 3w_1 + b \geq 1 & (4) \end{cases}$$

对约束条件计算可得：

$$(1) + (3) \Rightarrow w_1 \geq +1$$

$$(2) + (3) \Rightarrow w_2 \leq -1$$

由目标函数 $\frac{1}{2} \mathbf{w}^T \mathbf{w}$ 取最小值可得：

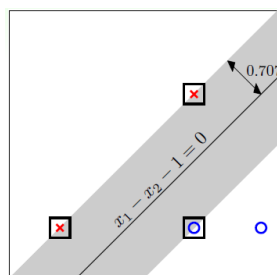
$$w_1 = 1, w_2 = -1$$

代入约束条件： $b = -1$

$$\hat{y} = g(\mathbf{x}) = \text{sign}(x_1 - x_2 - 1)$$

7.3 支撑向量机

为什么叫支撑向量机？



$$\mathbf{X} = \begin{bmatrix} 0 & 0 \\ 2 & 2 \\ 2 & 0 \\ 3 & 0 \end{bmatrix} \quad \mathbf{y} = \begin{bmatrix} -1 \\ -1 \\ +1 \\ +1 \end{bmatrix}$$

$$\min_{\mathbf{w}} \quad \frac{1}{2} \mathbf{w}^T \mathbf{w}$$

$$\text{Subject to} \quad y_n(\mathbf{w}^T \mathbf{x}_n + b) \geq 1, \text{ for all } n$$

优化得到的解为：

$$w_1 = 1, w_2 = -1, b = -1$$

$$g(\mathbf{x}) = \text{sign}(x_1 - x_2 - 1)$$

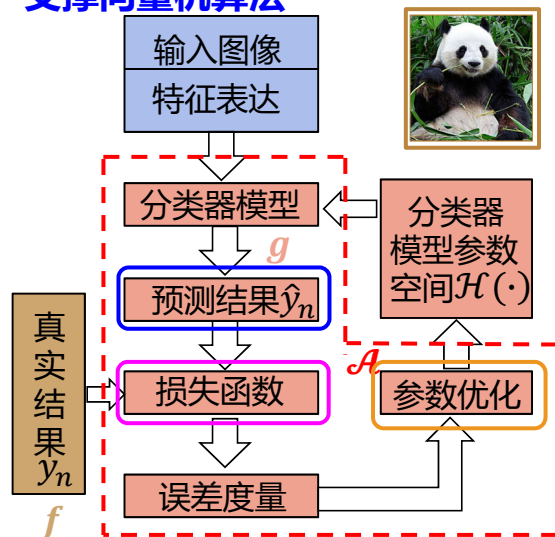
$$\text{margin}(\mathbf{w}) = \frac{1}{\|\mathbf{w}\|} = \frac{1}{\sqrt{2}}$$

- 分类面由边界上的样本确定，其他样本不起作用
- 边界上的样本被称为支撑向量(候选)

支撑向量机(SVM)—Support Vector Machine
----借助支撑向量学到间隔最大分类面

7.3 支撑向量机

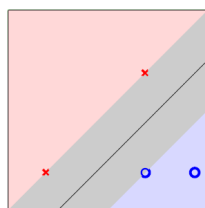
支撑向量机算法



$$\min_w \frac{1}{2} w^T w$$

$$\text{Subject to } y_n(w^T x_n + b) \geq 1, \text{ for all } n$$

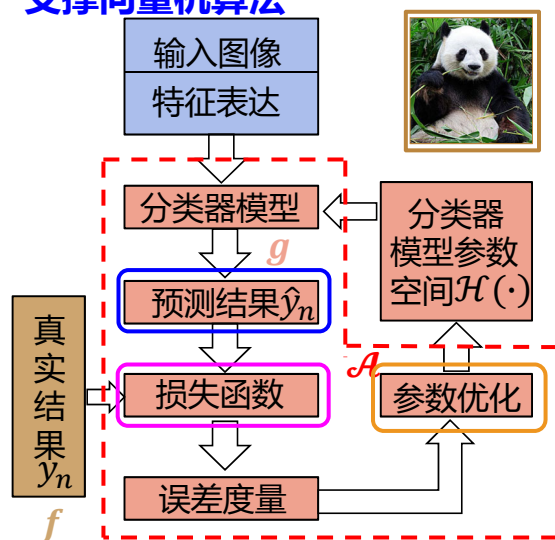
$$\hat{y}_n = \text{sign}(w^T x_n + b)$$



通过求解目标函数的最优解找到最大间隔作为分类面

7.3 支撑向量机

支撑向量机算法



$$\min_w \frac{1}{2} w^T w$$

$$\text{Subject to } y_n(w^T x_n + b) \geq 1, \text{ for all } n$$

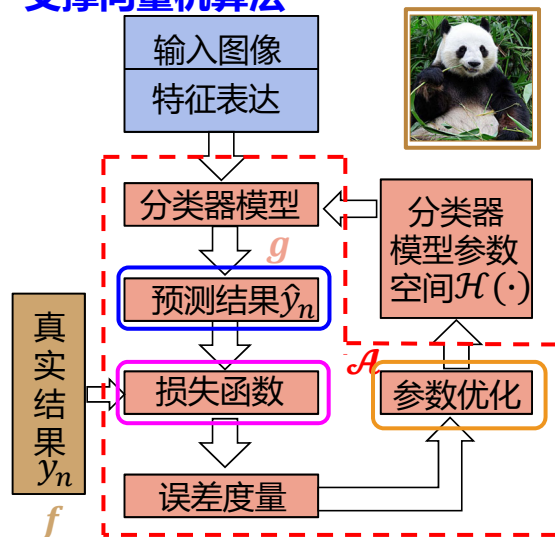
感知器算法的损失函数:

$$L_{in} = \sum_{n=1}^N \mathbb{I}[y_n \neq \hat{y}_n] = 0$$

$$\hat{y}_n = \text{sign}(w^T x_n + b)$$

7.3 支撑向量机

支撑向量机算法



$$\min_w \frac{1}{2} \mathbf{w}^T \mathbf{w}$$

$$\text{Subject to } y_n(\mathbf{w}^T \mathbf{x}_n + b) \geq 1, \text{ for all } n$$

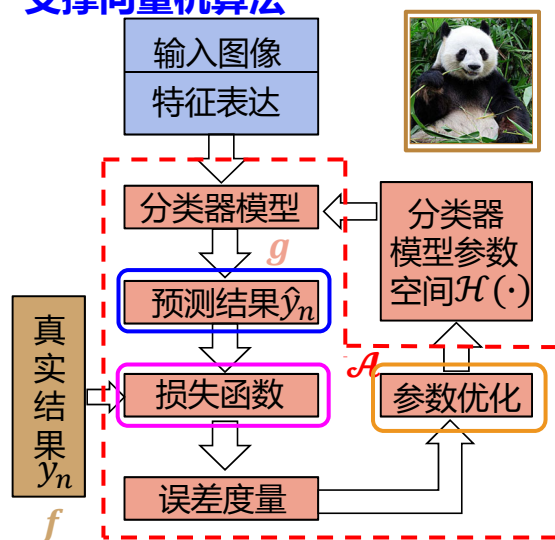
支撑向量机算法的损失函数?

$$L_{SVM} = ?$$

$$\hat{y}_n = \text{sign}(\mathbf{w}^T \mathbf{x}_n + b)$$

7.3 支撑向量机

支撑向量机算法



$$y_n(\mathbf{w}^T \mathbf{x}_n + b) \geq 1 \quad s_n = (\mathbf{w}^T \mathbf{x}_n + b)$$

$$y_n s_n \geq 1$$

$$1 - y_n s_n \leq 0$$

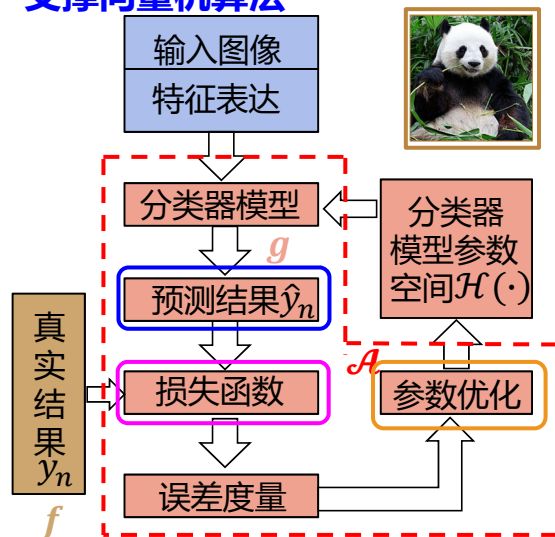
支撑向量机算法的损失函数?

$$L_{SVM} = ?$$

$$\hat{y}_n = \text{sign}(\mathbf{w}^T \mathbf{x}_n + b)$$

7.3 支撑向量机

支撑向量机算法



$$y_n(\mathbf{w}^T \mathbf{x}_n + b) \geq 1$$

$$y_n s_n \geq 1$$

$$1 - y_n s_n \leq 0$$

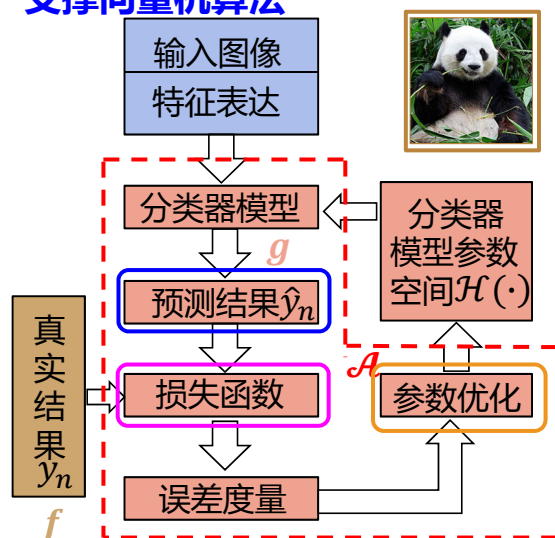
支撑向量机算法的损失函数

$$L_{SVM} = \max(0, 1 - y_n s_n) \quad \text{Hinge Loss}$$

$$\hat{y}_n = \text{sign}(\mathbf{w}^T \mathbf{x}_n + b)$$

7.3 支撑向量机

支撑向量机算法



$$y_n(\mathbf{w}^T \mathbf{x}_n + b) \geq 1$$

$$y_n s_n \geq 1$$

$$1 - y_n s_n \leq 0$$

支撑向量机算法的损失函数

$$L_{SVM} = \max(0, 1 - y_n s_n) \quad \text{Hinge Loss}$$

$$\frac{\partial L_{SVM}(\mathbf{w})}{\partial \mathbf{w}} = \mathbb{I}[1 - y_n(\mathbf{w}^T \mathbf{x}_n) \geq 0](-y_n \mathbf{x}_n) \quad (\text{增广后})$$

$$\hat{y}_n = \text{sign}(\mathbf{w}^T \mathbf{x}_n + b)$$

7.3 支撑向量机

梯度下降法实现支撑向量机

- 初始化权向量 w_0 **Stochastic Gradient Descent(SGD)**

- **for** $t = 0, 1, 2, \dots$ (t 代表迭代次数)

① 计算梯度: $\nabla L_{SVM}(w_t) = \frac{1}{B} \sum_{n=1}^B \llbracket 1 - y_n(w^T x_n) \geq 0 \rrbracket (-y_n x_n)$

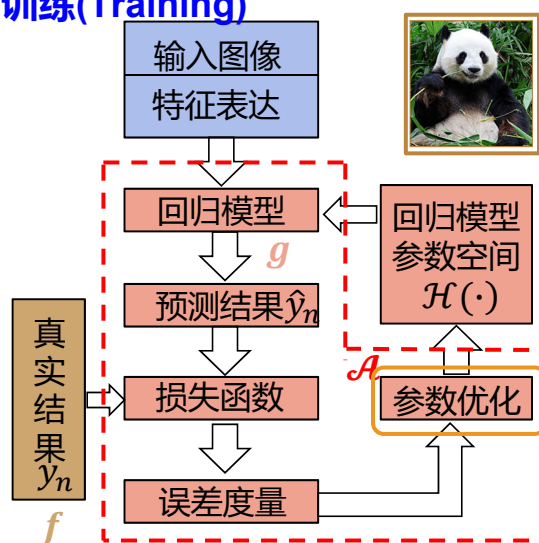
② 对权向量 w_t 进行更新: $w_{t+1} \leftarrow w_t - \eta \nabla L_{SVM}(w_t)$

...直到对任意 x_n 满足 $1 - y_n(w_{t+1}^T x_n) < 0$, 或者迭代足够多次数

返回最终的 w_{t+1} 作为学到的 g

3.3 梯度下降法 (引)

训练(Training)



随机梯度下降法(SGD):

$$\nabla L_{in}(w) = \sum_{n=1}^B (w^T x_n - y_n) x_n$$

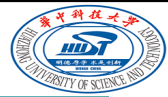
$$m_{i,t+1} = \lambda m_{i,t} + \eta \nabla L_{in}(w)$$

$$w_{i,t+1} \leftarrow w_{i,t} - m_{i,t+1}$$

不同损失函数
在梯度下降时
的速度不同

- 问题1: 学习率 η 的选择?
- 问题2: 梯度为0时, 是否收敛?
- 问题3: 训练样本数量大小的影响?
- 问题4: 损失函数的影响?

5.2 逻辑斯蒂回归损失（引）



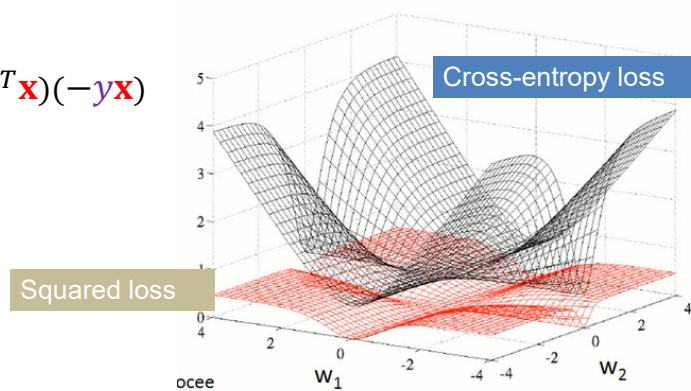
交叉熵损失与平方损失的梯度对比：

平方损失的梯度：

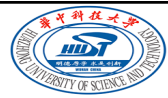
$$\nabla L_{in}(\mathbf{w}, \mathbf{x}, y) = 2(\theta(\mathbf{y}\mathbf{w}^T \mathbf{x}) - 1)\theta(\mathbf{y}\mathbf{w}^T \mathbf{x})(1 - \theta(\mathbf{y}\mathbf{w}^T \mathbf{x}))\mathbf{y}\mathbf{x}$$

交叉熵损失的梯度：

$$\nabla L_{in}(\mathbf{w}, \mathbf{x}, y) = \theta(-\mathbf{y}\mathbf{w}^T \mathbf{x})(-\mathbf{y}\mathbf{x})$$



7.3 支撑向量机



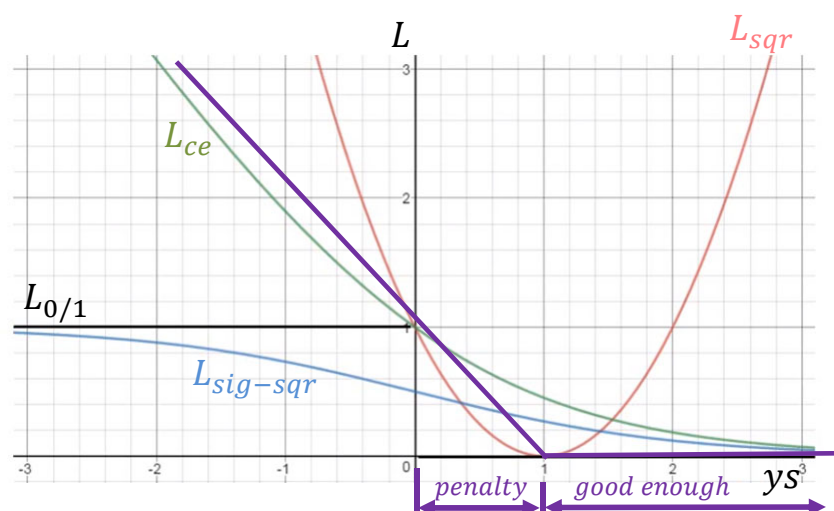
$$L_{0/1} = \mathbb{I}[\hat{y} \neq y]$$

$$L_{sqr} = (ys - 1)^2$$

$$L_{sig-sqr} = (\theta(ys) - 1)^2$$

$$L_{ce} = \ln(1 + \exp(-ys))$$

$$L_{SVM} = \max(0, 1 - ys)$$



7.3 支撑向量机

SVM的一般求解:

$$\min_{\mathbf{w}} \quad \frac{1}{2} \mathbf{w}^T \mathbf{w}$$

$$\text{Subject to} \quad y_n(\mathbf{w}^T \mathbf{x}_n + b) \geq 1, \text{ for all } n$$

- 一般情况下, “手工”求解不容易!
- 用梯度下降法? 如何处理约束条件?

SVM求解模型的特点:

- $(\mathbf{w}, b)$ 的目标函数是二次函数, 是凸函数!
- $(\mathbf{w}, b)$ 的约束条件是线性函数!

----二次规划(quadratic programming)问题!

二次规划(QP)
有成熟方便的办法求优化解!

7.3 支撑向量机

SVM的一般求解:

最佳的 $(\mathbf{w}, b) = ?$

$$\min_{\mathbf{w}} \quad \frac{1}{2} \mathbf{w}^T \mathbf{w}$$

$$\text{Subject to} \quad y_n(\mathbf{w}^T \mathbf{x}_n + b) \geq 1$$

$$\text{for } n = 1, 2, \dots, N$$

通过调用二次规划(QP)的求解函数就能得到SVM的最优解

二次规划(QP)的求解:

最佳的 $\mathbf{u} \leftarrow \text{QP}(\mathbf{Q}, \mathbf{p}, \mathbf{A}, \mathbf{c})$

$$\min_{\mathbf{u}} \quad \frac{1}{2} \mathbf{u}^T \mathbf{Q} \mathbf{u} + \mathbf{p}^T \mathbf{u}$$

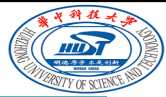
$$\text{Subject to} \quad \mathbf{a}_m^T \mathbf{u} \geq c_m$$

$$\text{for } m = 1, 2, \dots, M$$

$$\mathbf{u} = \begin{bmatrix} b \\ \mathbf{w} \end{bmatrix}, \quad \mathbf{Q} = \begin{bmatrix} 0 & \mathbf{0}_d^T \\ \mathbf{0}_d & \mathbf{I}_d \end{bmatrix}, \quad \mathbf{p} = \mathbf{0}_{d+1}$$

$$\mathbf{a}_n^T = y_n [1 \quad \mathbf{x}_n^T], \quad c_n = 1, \quad M = N$$

7.3 支撑向量机



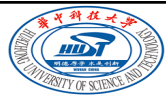
利用二次规划(QP)实现支撑向量机

- ① $\mathbf{Q} = \begin{bmatrix} 0 & \mathbf{0}_d^T \\ \mathbf{0}_d & \mathbf{I}_d \end{bmatrix}$, $\mathbf{p} = \mathbf{0}_{d+1}$, $\mathbf{a}_n^T = y_n [1 \quad \mathbf{x}_n^T]$, $c_n = 1$,
- ② $\begin{bmatrix} b \\ \mathbf{w} \end{bmatrix} \leftarrow \text{QP}(\mathbf{Q}, \mathbf{p}, \mathbf{A}, \mathbf{c})$
- ③ 返回最终的 b 和 $\mathbf{w}$ 作为学到的 $\mathbf{g}_{\text{SVM}}$

线性硬间隔SVM算法(Linear Hard-Margin SVM Algorithm)

- *Hard-Margin*: 没有任何样本会落入到“胖胖的”间隔区域里!
- *Linear*: 样本 $\mathbf{x}_n$ 是线性可分的! 如果不是线性可分? $\mathbf{z}_n = \Phi(\mathbf{x}_n)$

第七讲 线性支撑向量机 (Linear Support Vector Machine)



- 7.1 最大间隔分类面 (Large-Margin Separating Hyperplane)
直觉上对噪声更为鲁棒
- 7.2 标准的最大间隔问题 (Standard Large-Margin Problem)
最小化 $\mathbf{w}$ 的模长, 使得对所有样本都正确分类
- 7.3 支撑向量机 (Support Vector Machine)
利用二次规划QP获得最佳解